ENGINEERING DESIGN REPORT

Custom 3D-Printed Micro Quadrotor with Servo Payload Release

Drone Project — Redesigned Small-Scale Hobby/Engineering Build

Prepared for: University of Calgary — Engineering Student Hobby Project

Prepared by: Undergraduate Engineering Student (haseebsyed2006@gmail.com)

Date: September 4, 2026

Revision: B — adds a custom ESP32 RC link, replacing the FlySky TX/RX pair (see Design_Journal.md)

All values are labeled Manufacturer-specified, Calculated, Estimated, or Measured. No test, thrust, or flight result is fabricated — unmeasured KPIs are marked TARGET until actually tested.

Table of Contents

SECTION 2

Executive Summary

This report documents the redesigned scope of a small, affordable hobby quadcopter project: a custom 3D-printed 2.5-3 inch micro quadrotor with a servo-actuated payload-release mechanism. This design deliberately replaces an earlier, much larger autonomous-UAV concept (archived in _Archive_v1_Oversized_UAV/) that was well beyond the intended scope, budget, and timeline for this project.

The selected frame concept is a modular-arm design (Concept C of three compared), chosen for repairability during iterative testing. The propulsion system is a real, sourced 2S brushless setup (1103 11000KV motors, 65mm props, an AIO flight controller/ESC board) running Betaflight - no custom flight-stabilization firmware. A servo-actuated latch mechanism (SG90) delivers a 20-100 g payload.

RC control is a self-written link between two ESP32 boards (ESP-NOW wireless protocol, a custom SBUS encoder, and a failsafe watchdog on the onboard receiver) instead of a stock FlySky transmitter/receiver pair - a deliberate scope choice to demonstrate embedded programming alongside the mechanical work, at close to the same purchased cost. This is a real added engineering/schedule risk relative to a proven off-the-shelf receiver, and is documented honestly as such (Section 12).

Real, verified component pricing pushes the fully-capable build (meeting the stated 100-300 g mass / 20-100 g payload target) to approximately $155 CAD, above the original $100 CAD target - this is presented honestly as a real market constraint, not hidden. The user has indicated some willingness to stretch the budget; a cheaper brushed-motor fallback (~$125 CAD) is also documented, at reduced payload capacity.

Nothing has been physically built or tested yet. Every quantitative claim in this report is labeled Calculated, Estimated, or Manufacturer-specified - never a fabricated measurement. Target timeline is 3-8 weeks of part-time work.

Requirements

Project: Custom 3D-Printed Micro Quadrotor with Servo Payload Release Scale: Small hobby/engineering-student project, not a capstone-level UAV. This replaces the original requirements I wrote for a much bigger autonomous UAV concept; see ../_Archive_v1_Oversized_UAV/ for that version, kept only for reference.

1. Project Intent

Build a small, affordable, functional quadcopter, designing as much of the physical system as I reasonably can, then add a simple custom payload-delivery mechanism. The goal is real engineering depth (CAD, mechanical design, 3D printing, basic electronics, wiring, embedded systems, basic control systems, testing, iterative engineering) at a scale I can actually build and fly in 3-8 weeks of part-time work, not months.

Target claim this project should support:

> *"Designed and fabricated a lightweight custom quadrotor frame using CAD and additive manufacturing, integrating flight-control electronics and a servo-actuated payload delivery mechanism."*

2. Hard Constraints

Constraint	Value
Purchased hardware budget	$100 CAD strict target, with some flexibility to stretch if needed - see ../BOM/BOM.xlsx for why real component pricing pushes the fully-capable (brushless, 100-300 g class) build to ~$155 CAD, and the cheaper brushed-motor fallback that stays much closer to $100 CAD at reduced payload capacity
Vehicle size	2.5-5 inch propellers, small lightweight frame
Target all-up mass	~100-300 g (brushless path) if practical
Payload target	20-100 g, depending on actual lifting capacity
Timeline	3-8 weeks part-time
Team	Solo student
Fabrication access	3D printers, basic electronics tools, soldering equipment, CAD software, basic workshop tools, University of Calgary engineering resources, various sensors/batteries

3. Functional Requirements

- FR1: Custom-designed, 3D-printed frame, not a purchased ready-to-fly frame.
- FR2: IMU/gyro-based flight stabilization using existing flight-control firmware (Betaflight or equivalent). I'm not reimplementing flight-critical stabilization software from scratch.
- FR3: Manual RC flight control, no autonomous navigation required. The RC link itself is a custom ESP32-to-ESP32 link I write (ESP-NOW + a self-encoded SBUS output to the FC) instead of a stock transmitter/receiver pair, see ../Electrical/Electrical_Architecture.md §1a; this is a deliberate scope choice to demonstrate embedded programming, not a requirement to reimplement flight stabilization (FR2 still stands: Betaflight handles that).
- FR4: A custom-designed, 3D-printed, servo-actuated payload-release mechanism, manually triggered by the pilot.
- FR5: Basic wiring/electronics integration I assemble myself (frame, motors, ESC/FC, battery, servo).

4. Non-Functional Requirements

- NFR1: Feasible for one student within 3-8 weeks part-time.
- NFR2: Budget discipline: prioritize university inventory, then AliExpress/cheap suppliers, using real verified prices, never invented costs.
- NFR3: Documentation integrity: every number labeled Manufacturer-specified, Calculated, Simulated, Estimated, or Measured. No fabricated test, FEA, or flight results. Unmeasured KPIs marked TARGET until actually tested.

- NFR4: Reuse flight-critical firmware. Use an existing, proven flight controller/firmware (Betaflight, or a documented Arduino/ESP32/STM32 plus MPU6050-class alternative only if it's at least as reliable and doesn't eat weeks of development time) instead of writing my own stabilization control law.

5. Explicitly Out of Scope

GPS, LiDAR, Raspberry Pi/companion computer, computer vision, SLAM, optical flow, autonomous navigation, obstacle avoidance, expensive telemetry, custom PCB design, advanced ground station, large batteries, heavy payloads, carbon-fiber frame, any $500+ component. All of this was in scope for the archived v1 design; it's deliberately excluded here.

6. Success Criteria

1. Vehicle flies reliably under manual control.
2. Frame, motor mounts, and payload mechanism are original CAD/3D-printed designs with a documented, if simple, load case.
3. Payload-release mechanism reliably carries and releases a 20-100 g payload (brushless path) or a reduced payload appropriate to actual lifting capacity (brushed fallback).
4. A short, honest test record exists (../Testing/Test_Plan.xlsx) with TARGET values replaced by measured ones as testing happens.
5. The build is documented clearly enough to write 3-5 honest, unexaggerated resume bullets.

7. Open Questions

See the design-decision notes in ../Mechanical/Frame_Design.md and ../Electrical/Electrical_Architecture.md for the brushless-vs-brushed cost/capability trade-off, and ../README.md for what's still unconfirmed (exact university inventory availability, final budget ceiling, which frame concept to build first).

SECTION 4

Research

Real, web-researched small/budget quadcopter projects, replacing the original research aimed at a much larger autonomous UAV. Focus question: what is the best small, inexpensive quadcopter project that still has meaningful engineering design work?

	Project	Source	Relevance
1	Feather 3, Ultra-light FPV Toothpick Frame	MakerWorld (https://makerworld.com/en/models/1424214-feather-3-ultra-light-fpv-toothpick-frame)	Free, real 3D-printable 3-inch toothpick frame design: ~5 g printed frame, complete builds under 30 g, sized for standard 25.5x25.5 mm AIO boards. Directly usable as a starting-point reference for my own custom frame design (Concept A/B in Concept_Comparison.md).
2	"How to build a cool & cheap 3D printed micro drone"	Prusa blog (https://blog.prusa3d.com/how-to-build-a-3d-printed-micro-drone_29310/)	Prusa's own build guide for a cheap, 3D-printed micro FPV drone, a real, credible source (printer manufacturer, not a random blog) confirming this class of build is a well-trodden, achievable hobby project.
3	Ultralight 3 Inch (75 g)	RotorBuilds (https://rotorbuilds.com/build/18341)	A real documented build log of a 75 g, 3-inch ultralight, close to my upper target mass before payload, useful as a direct mass/component benchmark.
4	The FPV Shopping List, Toothpick FPV Drones & Parts	FPV Know-It-All (Joshua Bardwell) (https://www.fpvknowitall.com/fpv-shopping-list-toothpick-fpv-drones-and-parts/)	A well-known, credible FPV educator's curated parts list for toothpick-class builds, used to cross-check my own component category choices (FC/ESC/motor/prop/battery) against community-standard practice.
5	Toothpick and Ultralight 3-Inch FPV Drone Build Guide 2026	UAVMODEL Insights (https://blog.uavmodel.com/toothpick-and-ultralight-3-inch-fpv-drone-build-guide-2026-parts-list-and-assembly/)	Current (2026) build guide with parts list and assembly steps for exactly this class of drone. Confirms toothpick-class builds stay under ~250 g AUW, and that 3S with 4500-5000KV motors is a common "first toothpick" recommendation (informed, but not directly copied, my own motor/battery choice given the tighter budget).
6	Kabab-Toothpick-3 (TP3) Frame	FPVCycle (https://fpvcycle.com/products/kabab-toothpick-3-tp3-frame)	A real commercial 3-inch toothpick frame product, used as a cost/design benchmark (what a "finished" version of this frame class sells for) against which my own 3D-printed design is compared in Concept_Comparison.md.
7	Payload Dropper Mechanism	Printables (Aman_Bhatt) (https://www.printables.com/model/1326088-payload-dropper-mechanism)	Free, real, downloadable STL for a drone payload-dropper mechanism, a direct real-world precedent for Mechanical/Payload_Mechanism.md.
8	DIY Drones Deliver The Goods With Printed Release	Hackaday, 2025-01-22 (https://hackaday.com/2025/01/22/diy-drones-deliver-the-goods-with-printed-release/)	Real Hackaday coverage of a 3D-printed servo-actuated drone payload-release mechanism, confirming this is a credible, previously-built approach, not a novel/unproven idea.
9	DIY Drone Payload Release Mechanism: Step-by-Step Build	Zbotic (https://zbotic.in/diy-drone-payload-release-mechanism-step-by-step-build/)	Build-guide-level detail on servo torque/mechanism-force sizing for a payload release latch, directly informs the servo selection and force analysis in Mechanical/Payload_Mechanism.md.

Synthesis

Synthesis

The dominant, well-precedented archetype for a small, cheap, engineering-rich quadcopter is the "toothpick" class: a 2.5-3 inch prop, 3D-printed frame (often carbon-fiber-tube-reinforced), a single AIO flight-controller/ESC board, small brushless motors, and a small 2S LiPo, commonly landing in the 30-250 g AUW range depending on motor/prop size. That's a good match for this project's 100-300 g target once a payload mechanism and slightly larger motors get added. Servo-actuated payload-release mechanisms (hook, trapdoor, or latch, SG90-class servo) are a real, repeatedly-built, low-risk addition to this airframe class (sources 7-9), exactly the "interesting feature" I wanted without needing any flight-software changes.

What this rules out relative to the original (archived) research: none of the credible small-build sources above use GPS, LiDAR, companion computers, or autonomous navigation, which confirms that leaving those out is consistent with how this class of project actually gets built by the hobby/maker community, not a corner being cut.

SECTION 5

Concept Comparison

Comparison Table

Criterion	A: Traditional X	B: Compact Integrated	C: Modular Arm
Mass	Best	Worst	Good
Stiffness	Good	Best	Good (joint-dependent)
Print time	Best	Worst	Good
Material usage	Best	Worst	Good
Ease of assembly	Good	Best	Fair
Repairability	Good	Worst	Best
Motor mounting	Best (standard)	Fair (tight spacing)	Best (standard)
Payload integration	Fair (add-on)	Good (if planned right the first time)	Fair (add-on)

SECTION 6

Selected Design

Selected Concept: C, Modular Arm Design

Selected over Concept A (traditional X) by a narrow margin, and clearly over Concept B (compact integrated body). Reasoning:

1. Repairability matters most for a first hobby build. My first flights are the most likely to end in a hard landing or crash. A design where a single snapped arm costs one small reprint and a few minutes of reassembly (Concept C) is far more forgiving, and far cheaper to iterate on, than one where a crash means reprinting the whole frame (Concept B) or, to a lesser extent, Concept A's continuous arm-to-plate joint (repairable by reprinting the whole plate, but doesn't allow single-arm replacement).
2. It keeps nearly all of Concept A's mass/print-time/material advantages. The modular joint adds only a small mass/print-time penalty, not Concept B's much larger one.
3. It's a real engineering trade-off decision (stiffness/joint-design vs. repairability) that makes for a genuinely good interview talking point, more so than just picking the "obviously lightest" option without justification.
4. It matches a real, credible precedent. The modular/repairable-by-design philosophy is explicitly used by real projects referenced in prior research, so this isn't a novel, unvalidated idea.

Concept B is rejected as the frame concept: for a small hobby project meant to be iterated on quickly and cheaply within a 3-8 week timeline, a design where any meaningful crash requires a full-frame reprint works against the whole point of being fast, cheap, and low-friction to iterate on.

Full frame dimensions, arm-joint fastener sizing, and the payload-mount interface are in ../Mechanical/Frame_Design.md and ../Mechanical/CAD_Architecture.md.

SECTION 7

System Architecture

The system is deliberately simple (per the brief's Section 4): a single flight-controller/ESC board running Betaflight handles all stabilization and motor control; a self-written ESP32-to-ESP32 RC link (ESP-NOW wireless, custom SBUS output) provides manual pilot control in place of a stock receiver; a payload servo, triggered from a spare AUX channel carried over that link, handles payload release. No companion computer, no autonomy stack, no custom PCB.

Block Diagram

Battery -> FC/ESC (AIO board) -> 4x brushless motors (thrust/attitude control). Handheld TX ESP32 --ESP-NOW (wireless)--> onboard RX ESP32 -> FC/ESC (SBUS, pilot control). FC/ESC -> Payload servo (PWM, AUX-channel triggered release). Full labeled wiring diagram: Section 13.

Subsystem Ownership

Subsystem	Owner document
Mechanical structure (frame, arms, payload mount)	Mechanical/CAD_Architecture.md, Frame_Design.md
Propulsion & structural calculations	Mechanical/Frame_Design.md
Payload mechanism	Mechanical/Payload_Mechanism.md
Manufacturing	Mechanical/Manufacturing_Plan.md
Simple FEA	Mechanical/FEA_Plan.md
Electronics/wiring	Electrical/Electrical_Architecture.md, Wiring_Diagram.pdf
RC-link firmware (ESP-NOW, SBUS, failsafe)	Electrical/Electrical_Architecture.md §1a
Bill of materials	BOM/BOM.xlsx
Testing	Testing/Test_Plan.xlsx

2. Thrust Requirement (Calculated target, NOT a claimed achieved thrust)

Using a mid-range payload target of 50 g (within the 20-100 g range) for baseline sizing:

- Total system mass = 90 g (bare) + 50 g (payload) = 140 g
- Weight = 0.140 kg x 9.81 m/s² = 1.37 N
- Target thrust-to-weight ratio = 2:1
- Required total thrust = 2 x 1.37 N ~ 2.75 N ~ 280 gf, i.e. ~ 70 gf per motor (4 motors)

This is the required number, not a claimed capability. I couldn't find a real, independently-verified thrust-in-grams curve for the specific 1103 11000KV + 65mm prop + 2S combination (one bench-test source, fishpepper.de, turned up but its actual data tables weren't accessible in this pass, so it's flagged as NOT FOUND, not estimated). What is real and sourced: this exact motor/prop/cell-count combination is the standard, widely-used configuration for 2S toothpick/whoop-class builds in this weight class (Existing_Projects.md, sources 1-6), which gives reasonable qualitative confidence that ~70 gf/motor is achievable well under full throttle. The actual achieved thrust has to be confirmed by Test 6 (Static Thrust Test) in ../Testing/Test_Plan.xlsx before I trust it for real flight. If bench testing shows the motors fall short of ~70 gf/motor at a reasonable throttle percentage, the payload target needs to come down accordingly. That's recorded here as an open item, not silently assumed to work.

Component Selection

1. Component Selection (Recommended / Brushless path)

Role	Selected component	Key specs	Price (USD)	Source
Flight controller + ESC	2S F4 AIO FC + Brushless ESC (no RX)	STM32F4, OSD, SmartAudio, built-in 4-in-1 ESC	$22.99	AliExpress (https://www.aliexpress.com/item/32969058968.html)
RC link — onboard RX	ESP32-C3 Super Mini	RISC-V, 2.4GHz WiFi, custom firmware: ESP-NOW in -> SBUS out (inverted UART) to the FC	~$3.50-4.80	AliExpress (https://www.aliexpress.com/item/1005005319963906.html)
RC link — handheld TX	ESP32-WROOM-32 DevKit	Dual-core, 2.4GHz WiFi, custom firmware: reads sticks/switches -> ESP-NOW out	$2.20	AliExpress (https://www.aliexpress.com/item/1005003145871431.html)
RC link — sticks	KY-023 dual-axis joystick module x2	4 axes total (throttle, yaw, pitch, roll)	$0.60 each	AliExpress (https://www.aliexpress.com/item/1882818018.html)
RC link — switches	Momentary/toggle switch x2	Arm switch, payload-release trigger (AUX channels)	~$0.50 each	Various, see ../BOM/BOM.xlsx
RC link — TX power	USB power bank (5V)	Powers the handheld TX over USB, assumed already owned	already owned / UNIVERSITY	not counted in purchase cost unless confirmed unavailable
Motors (x4)	1103 11000KV brushless	~3.3-3.55 g each, 2S-rated, 1.5mm shaft	$50.99 (4-pack)	Pyrodrone (https://pyrodrone.com/products/betafpv-1103-11000kv-2s-brushless-motor-4pcs)

Role	Selected component	Key specs	Price (USD)	Source
Propellers (x4 + spares)	Gemfan 65mm 2-blade	1mm/1.5mm shaft, set of 8 (4 spare)	$4.49	GetFPV (https://www.getfpv.com/gemfan-65mm-micro-propellers-1mm-shaft-set-of-8.html)
Battery	2S 450mAh HV LiPo (Turnigy BoltX 80C)	7.6V nominal, XT30	$5.99	HobbyKing (https://hobbyking.com/en_us/turnigy-nano-tech-300mah-2s-35-70c-lipo-pack.html)-class listing, see ../BOM/BOM.xlsx for exact SKU
Payload servo	SG90 9g micro servo	1.8 kg·cm stall torque	~$3.00	AliExpress (bulk-pack pricing, see Payload_Mechanism.md)

Total purchased (Recommended, brushless): see ../BOM/BOM.xlsx, real-priced total is ~ $155 CAD, above my original $100 CAD target. This is a deliberate, documented trade-off, not a budget overrun I'm glossing over: real brushless FPV micro-motor pricing (~$51 USD for 4 motors, confirmed across both a name-brand retailer and a generic AliExpress listing at nearly identical prices) is simply the real cost floor for hardware that meets the 100-300 g / 20-100 g payload target. I've got room to stretch the budget somewhat, so I'm going with this. The Minimum tier below stays much closer to $100 CAD by accepting reduced payload capacity instead. Swapping the FlySky TX/RX pair for the custom ESP32 link is essentially cost-neutral, actually about $2 CAD cheaper (~$11.67 CAD combined for the RX/TX ESP32s, joysticks, and switches, vs. the ~$13.89 CAD the FlySky receiver alone cost, see ../BOM/BOM.xlsx), it's a scope decision about demonstrating embedded programming, not a budget decision.

1a. Custom RC Link (ESP32, replaces FlySky TX/RX)

I'm replacing the stock FlySky transmitter/receiver pair with a self-written RC link between two ESP32 boards. The reason is deliberate: it turns "buy a receiver" into an actual embedded-programming deliverable (packet protocol, SBUS generation, failsafe logic) that fits the project's stated embedded-systems goal (../Requirements/Requirements.md §1), at close to the same purchased cost as the FlySky pair it replaces. This is a real added engineering/schedule risk, not a drop-in swap — a commercial receiver already has proven failsafe and RF behavior; here I have to build and validate that myself before it's trusted in the air. Flagged honestly, not glossed over.

Link: ESP-NOW (built into the ESP32 WiFi radio, connectionless, no router/pairing needed, low latency, typically ~2-5 ms one-way).

TX firmware (handheld ESP32):

- Reads 4 analog stick axes (throttle, yaw, pitch, roll) + 2 switch inputs (arm, payload release) from GPIO/ADC.
- Packs them into a small struct (channel values + a sequence number + a checksum) and broadcasts it via ESP-NOW at ~50-100 Hz.

RX firmware (onboard ESP32):

- Receives ESP-NOW packets, validates the checksum, updates a local channel-value table.
- Independently, on its own fixed timer (not tied to packet arrival), builds a standard 25-byte SBUS frame from the current channel-value table and writes it out over UART to the FC, at the ~9-14 ms interval Betaflight expects. Decoupling the SBUS timer from RF packet timing keeps the signal to the FC steady even if individual radio packets are late or dropped.
- Failsafe watchdog: if no valid ESP-NOW packet has been received within a timeout (~200-300 ms), the RX firmware sets the SBUS frame's failsafe flag (and/or forces throttle to a safe low value) instead of holding the last good command, the same behavior a commercial receiver provides, but here it has to be explicitly written and bench-tested

(Test 5, ../Testing/Test_Plan.xlsx), not assumed.

SBUS signal inversion: standard SBUS is an inverted 100000-baud UART signal. The RX ESP32's UART peripheral can generate this directly (hardware signal-invert option exposed by the Arduino-ESP32 core's HardwareSerial.begin()), so no separate inverter circuit should be needed. Contingency: if the FC doesn't accept the ESP32's inverted signal cleanly, a simple NPN-transistor inverter (~$0.50 CAD in parts, a transistor + two resistors, not separately itemized in the BOM) is a documented fallback.

Known limitation vs. a commercial RC system, stated honestly: ESP-NOW is a WiFi-radio protocol, not a purpose-built RC link. It doesn't have the frequency-hopping/diversity/interference-hardening that AFHDS2A (FlySky) or dedicated systems like ExpressLRS have, and its realistic range with stock antennas is on the order of tens to a couple hundred meters, well short of a commercial system's typical range. That's an acceptable trade for a hobby build flown at close range in a controlled test area, but it's a real limitation, not a hidden one, and range/failsafe behavior must be bench-verified (walk-away failsafe test) before any flight test, not assumed to match FlySky's proven behavior.

SECTION 9

Mechanical Design

Concept: modular-arm 2.5-3 inch quadrotor with servo payload release (see ../Research/Concept_Comparison.md). Written package-agnostic, SolidWorks, Fusion 360, Onshape, whatever works.

1. Assembly Tree

```
Drone Assembly
%
% % %  Frame
%    % % %  Center Hub (electronics deck)
%    % % %  Arm x 4 (identical part, bolt-on)
%    % % %  Motor Mount x 4 (integral to arm tip, or separate, see §2)
%
% % %  Electronics Mount
%    % % %  FC/ESC standoff mount (soft-mount if vibration is an issue, see §3)
%
% % %  Battery Mount
%    % % %  Strap slot / friction-fit tray on top or bottom deck
%
% % %  Landing Gear
%    % % %  4 simple printed legs/feet (or skip entirely; a light toothpick-class build often lands fine on the frame
itself, and a small standoff foot is cheap insurance for the payload mechanism's ground clearance)
%
% % %  Payload Mechanism
    % % %  Servo Mount
    % % %  Latch/Hook Arm
    % % %  Payload Holder (see `Payload_Mechanism.md` for full design)
```

2. Key Dimensions & Interfaces (Estimated starting values, refine once CAD begins)

Feature	Value	Note
Prop size	65 mm (2.5")	Matches the selected 1103 11000KV motor/ prop combo
Motor-to-motor diagonal	~110-120 mm	Standard toothpick-class spacing for 65 mm props with safe clearance
Arm width	~6-8 mm	Estimated starting point, narrow enough to be light, wide enough for M2 bolt clearance at the hub joint
Arm thickness	3-4 mm	Print-oriented flat on its widest face for best layer-adhesion direction under bending (see Manufacturing_Plan.md)
Hub thickness	2-3 mm	Enough for M2 heat-set inserts without breaking through
Motor mounting pattern	Per 1103 datasheet, confirm exact bolt circle from the motor at CAD time, not assumed	Typical micro-motor mounts use a small M1.4-M2 bolt circle
FC/ESC mounting	25.5x25.5 mm standard AIO mounting holes (matches the Feather 3 frame reference and most whoop/toothpick AIO boards)	Confirm against the actual board purchased

3. Vibration Consideration

Even at this small scale, a rigid FC mount directly on the frame can pick up motor vibration into the gyro. A simple compliant mount (thin TPU standoffs, or foam tape between the FC and the deck) is cheap insurance. This is optional at this scale (unlike the archived v1 design, where it was a first-class requirement for GPS-denied position hold), since Betaflight's gyro filtering generally tolerates a reasonable amount of vibration on a build this light. I'll add it if bench testing (Test 4, Gyro/IMU validation) shows noisy gyro data.

4. Fasteners & Tolerances

- M2 hardware throughout (bolts, heat-set inserts), appropriately scaled for a sub-100g frame. M3 would be oversized and heavy for a build this small.
- Heat-set brass inserts at the arm-to-hub joint (the whole point of the modular design, per ../Research/Concept_Comparison.md) and at the payload-mechanism servo mount.
- Printed clearance tolerance: ±0.1-0.15 mm as a starting point on a well-calibrated desktop FDM printer, tighter than the larger v1 design needed since these are much smaller features.

5. Purchased Components as Reference Geometry

For fit-checking (not full detailed models, that would be overkill at this scale): block out the FC/ESC board footprint, motor bolt pattern, battery outline, and servo body from their real dimensions once purchased/confirmed, rather than guessing.

SECTION 10

Frame Calculations

1. Weight Budget (Estimated bottom-up mass, brushless/Recommended path)

Component	Mass	Basis
Frame (modular arms + hub, 3D printed)	10 g	Estimated, no CAD/print exists yet; benchmarked against the real Feather 3 frame (~5 g bare X-frame, source 1 in Existing_Projects.md), plus a margin for the modular arm-joint hardware bosses
Flight controller + ESC (2S F4 AIO combo)	6 g	Estimated, typical mass for this board class; exact figure not published on the listing found
Onboard RX (ESP32-C3 Super Mini)	3 g	Estimated, typical published mass for this compact board class (~22x18mm bare board), not independently re-verified this session
Motors (4x 1103 11000KV)	14 g	Manufacturer-specified, ~3.3-3.55 g each across multiple real listings (BetaFPV (https://betafpv.com/products/1103-brushless-motors), cross-checked against AliExpress/T-Motor listings)
Propellers (4x 65mm)	4 g	Estimated, small 2-blade micro props are typically ~1 g each
Battery (2S 450mAh HV, e.g. Turnigy BoltX)	28 g	Estimated, typical mass for this capacity/chemistry class, not independently re-verified this session
Payload-release servo (SG90)	9 g	Manufacturer-specified, the SG90's ~9 g mass is one of its best-documented specs
Payload mount/latch hardware (printed)	5 g	Estimated
Wiring, connectors, heat-shrink	5 g	Estimated
Total bare AUW (no payload)	~ 84 g	Sum, rounded to 90 g for margin

This lands comfortably within the 100-300 g target band once a payload is added, and lines up with real precedent (source 5 in Existing_Projects.md: toothpick-class builds "stay under ~250 g AUW"; source 3: a documented real 75 g 3-inch build).

2. Thrust Requirement (Calculated target, NOT a claimed achieved thrust)

Using a mid-range payload target of 50 g (within the 20-100 g range) for baseline sizing:

- Total system mass = 90 g (bare) + 50 g (payload) = 140 g
- Weight = 0.140 kg x 9.81 m/s² = 1.37 N
- Target thrust-to-weight ratio = 2:1
- Required total thrust = 2 x 1.37 N ~ 2.75 N ~ 280 gf, i.e. ~ 70 gf per motor (4 motors)

This is the required number, not a claimed capability. I couldn't find a real, independently-verified thrust-in-grams curve for the specific 1103 11000KV + 65mm prop + 2S combination (one bench-test source, fishpepper.de, turned up but its actual data tables weren't accessible in this pass, so it's flagged as NOT FOUND, not estimated). What is real and sourced: this exact motor/prop/cell-count combination is the standard, widely-used configuration for 2S toothpick/whoop-class builds in this weight class (Existing_Projects.md, sources 1-6), which gives reasonable qualitative confidence that ~70 gf/motor is achievable well under full throttle. The actual achieved thrust has to be confirmed by Test 6 (Static Thrust Test) in ../Testing/Test_Plan.xlsx before I trust it for real flight. If bench testing shows the motors fall short of ~70 gf/motor at a reasonable throttle percentage, the payload target needs to come down accordingly. That's recorded here as an open item, not silently assumed to work.

3. Battery (Estimated, pending bench confirmation)

- Battery: 2S 450 mAh HV LiPo (e.g. Turnigy BoltX 80C, 7.6 V nominal), Manufacturer-specified capacity/voltage, Estimated mass (see §1).

- Current draw at hover-equivalent throttle: NOT FOUND / not calculable from reliable data this session. The only power figures I found for this motor class (peak power/current ratings synthesized from listings) were internally inconsistent between sources, so I'm not using them here rather than risk presenting a wrong number as fact.
- Flight time: TARGET, to be measured in Test 7 (Hover Test). Community experience with this general build class (toothpick/whoop, 2S, small LiPo) typically falls in a several-minutes-per-charge range, but that's not claimed here as a design value, only as a rough sanity-check range to be confirmed or corrected by measurement.

4. Simple Structural Calculations

Arm loading (Calculated, preliminary)

Each arm carries its motor's reaction thrust. Using the required-thrust figure above (70 gf/motor ~ 0.69 N) as the working design load (not the motor's true max, which is unknown pending Test 6):

- Design load per arm ~ 0.69 N, applied at the arm tip (motor mount), reacted at the hub joint.
- A conservative preliminary target safety factor of >= 3 against this load is set for the arm design, consistent with standard practice for small FDM-printed structural parts (layer-adhesion anisotropy makes a lower margin risky). This is a design target, to be confirmed by the simple FEA study in FEA_Plan.md once CAD geometry exists.
- Because the true achievable motor thrust is unverified (§2), I'll also check the arm against a higher bounding case once Test 6 data exists, e.g. the motor's actual measured max thrust, not just the ~70 gf design target, since a crash/prop-strike event can briefly load the arm well above steady hover thrust.

Motor mount (Calculated, preliminary)

Concentrated load transfer from the motor's 4-hole (or similar) bolt pattern into the arm tip, same design load as above (0.69 N design case, re-checked once measured thrust data exists).

Payload attachment (Calculated, preliminary)

The payload mount must support the payload's static weight (0.20-1.00 N for a 20-100 g payload) plus a dynamic/landing-impact allowance. A conservative 3x static weight impact-load design case is used (matching common small-UAV landing-load practice): 0.6-3.0 N depending on the actual payload chosen.

Full FEA (a single simple static load case, kept deliberately small) is in FEA_Plan.md.

Simple FEA

The idea here: if FEA is practical, do a small FEA study of the custom frame, not a giant project. A simple static load case is enough. This document defines that one simple study, not a multi-load-case aerospace-style analysis (see the archived v1 FEA_Plan.md for what that looked like, and why it was overkill for this project's scale).

1. What Gets Analyzed

One part, one load case: the arm, under the design thrust load.

- Part: a single modular arm (from ../Research/Concept_Comparison.md's selected Concept C).
- Load case: 0.69 N applied at the motor-mount end (the required-thrust design load from Frame_Design.md §2), fixed constraint at the hub-joint bolt holes.
- Material: PETG, standard published FDM material properties for the slicer/filament used (use the filament manufacturer's datasheet tensile strength, which counts as Manufacturer-specified, rather than a generic "PETG" textbook value, since real FDM part strength varies a lot by print settings).

2. Target

Factor of safety >= 3 against the 0.69 N design load, given FDM print-quality uncertainty (same conservative logic as the archived v1 design, just applied to a much smaller, simpler load case here).

3. Method

- A single static linear FEA run in whatever CAD package I'm using (SolidWorks Simulation, Fusion 360 Simulation, or similar built-in tool, no separate FEA software needed at this scale).

- Default mesh is fine for a part this small and simple; a single mesh-refinement check at the bolt-hole stress concentration is good practice but not mandatory at this scale (unlike the archived v1's more rigorous mesh-convergence requirement).

4. What Happens With the Result

- If FoS >= 3: proceed to printing as designed.
- If FoS < 3: thicken the arm slightly or add a rib, a fast, cheap iteration at this part size (see Manufacturing_Plan.md's ~10-15 minute per-arm print time).
- The result gets labeled Simulated in the documentation, and since the true achievable thrust is still pending Test 6 (Frame_Design.md §2), I'll spot-check the arm again once real thrust data exists in case the design load needs revising upward.

5. Status

Not started. No CAD geometry exists yet. This will run once frame CAD is modeled, as a same-day check, not a separate project phase.

SECTION 11

Payload Mechanism

This is the project's "interesting feature," a small, custom-designed, 3D-printed, servo-actuated payload release, manually triggered by the pilot. No autonomous release logic is needed.

1. Mechanism Selection

Real precedents researched (../Research/Existing_Projects.md, sources 7-9): servo-actuated hook release, trapdoor, and latch mechanisms are all real, previously-built, low-complexity approaches.

Selected: servo-actuated rotating latch/hook. Reasoning:

- Simpler to print and tune than a trapdoor (fewer moving/sealing surfaces).
- Lower mechanism-force uncertainty than a claw/gripper (a claw needs to grip a range of payload shapes reliably; a hook just needs to release a fixed attachment point).
- Directly matches the Printables/Hackaday/Zbotic precedents (sources 7-9), so this is a validated approach, not a novel unproven mechanism.

Mechanism sequence: the payload hangs from a printed hook/cradle attached to a servo horn. At rest, the servo holds the hook in the closed/latched position (payload secured). On pilot command (a spare RC channel/switch mapped in Betaflight, or a simple direct servo-signal trigger), the servo rotates the horn, opening the hook and releasing the payload by gravity.

2. Servo Selection & Force Analysis (Calculated)

- Selected servo: SG90 (9 g, ~1.8 kg·cm / ~1.6 lb·in stall torque, Manufacturer-specified, one of the most well-documented micro servo specs out there).
- Payload range: 20-100 g (0.196-0.981 N weight force).
- Moment arm: for a hook mounted close to the servo horn's pivot (~15-20 mm arm), required holding torque:
- At 100 g payload, 20 mm arm: T = F x r = 0.981 N x 0.020 m = 0.0196 N·m ~ 0.20 kg·cm
- The SG90's rated ~1.8 kg·cm stall torque gives a safety factor ~ 9x against the worst-case (100 g) payload at this moment arm, comfortable margin, and it lines up with the Zbotic reference's own sizing logic (source 9: "SG90 can hold ~900g of hatch plus payload force" at a 20mm arm, so this build's worst case of 100g is well within that same servo's demonstrated capability window).
- Conclusion: the SG90 is adequately sized with a large margin even at the top of the payload range. No need to step up to a heavier MG90S/MG996R servo, which would just add unnecessary mass/cost.

3. Design Details

Item	Design decision
Hook/latch material	PETG (impact resistance for repeated release cycles; see Manufacturing_Plan.md)
Payload cradle	A simple printed loop/cup the payload hangs from or sits in, sized to the actual payload item (open item, depends on what's actually being "delivered," e.g. a small printed test mass)
Servo mount	Printed bracket, M2 bolted to the frame hub, positioned so the release swing arc doesn't intersect props or landing legs
Trigger	A dedicated switch on the handheld ESP32 TX, carried as an AUX channel over the custom RC link (../Electrical/Electrical_Architecture.md §1a) and mapped to a servo output in Betaflight (servo mixer/AUX assignment) same as it would be with a stock receiver. The Betaflight-side mixer setup needs no custom firmware; getting that AUX channel there over the custom link does, see the RC-link firmware notes
Release testing	Bench-tested repeatedly (Test 8, Payload Mechanism Test) before any flight test. Release reliability is measured (release success rate over N trials), not assumed

4. Safety Factor Summary

Load case	Force	Servo capability	Safety factor
Max payload (100 g), 20 mm arm	0.0196 N·m	0.177 N·m (1.8 kg·cm)	~ 9x

Load case	Force	Servo capability	Safety factor
Min payload (20 g), 20 mm arm	0.0039 N·m	0.177 N·m	~ 45x (over-margined at the low end, which is fine since the servo is sized for the worst case)

5. Open Items

- Exact payload item(s) to be delivered: not decided yet, I'll assume a small printed test mass (e.g. a 3D-printed calibration weight) for initial testing, keeping with the "payload should be very small" idea.
- Exact moment arm depends on final CAD geometry. The 20 mm figure above is a reasonable placeholder, to be confirmed once the servo mount is modeled.

SECTION 12

Electrical Architecture

Concept: small modular-arm 2.5-3 inch quadrotor with servo payload release. Keeping it simple: no GPS, no companion computer, no custom PCB, no advanced telemetry.

1. Component Selection (Recommended / Brushless path)

Role	Selected component	Key specs	Price (USD)	Source
Flight controller + ESC	2S F4 AIO FC + Brushless ESC (no RX)	STM32F4, OSD, SmartAudio, built-in 4-in-1 ESC	$22.99	AliExpress (https://www.aliexpress.com/item/32969058968.html)
RC link — onboard RX	ESP32-C3 Super Mini	RISC-V, 2.4GHz WiFi, custom firmware: ESP-NOW in -> SBUS out (inverted UART) to the FC	~$3.50-4.80	AliExpress (https://www.aliexpress.com/item/1005005319963906.html)
RC link — handheld TX	ESP32-WROOM-32 DevKit	Dual-core, 2.4GHz WiFi, custom firmware: reads sticks/switches -> ESP-NOW out	$2.20	AliExpress (https://www.aliexpress.com/item/1005003145871431.html)
RC link — sticks	KY-023 dual-axis joystick module x2	4 axes total (throttle, yaw, pitch, roll)	$0.60 each	AliExpress (https://www.aliexpress.com/item/1882818018.html)
RC link — switches	Momentary/toggle switch x2	Arm switch, payload-release trigger (AUX channels)	~$0.50 each	Various, see ../BOM/BOM.xlsx
RC link — TX power	USB power bank (5V)	Powers the handheld TX over USB, assumed already owned	already owned / UNIVERSITY	not counted in purchase cost unless confirmed unavailable
Motors (x4)	1103 11000KV brushless	~3.3-3.55 g each, 2S-rated, 1.5mm shaft	$50.99 (4-pack)	Pyrodrone (https://pyrodrone.com/products/betafpv-1103-11000kv-2s-brushless-motor-4pcs)
Propellers (x4 + spares)	Gemfan 65mm 2-blade	1mm/1.5mm shaft, set of 8 (4 spare)	$4.49	GetFPV (https://www.getfpv.com/gemfan-65mm-micro-propellers-1mm-shaft-set-of-8.html)
Battery	2S 450mAh HV LiPo (Turnigy BoltX 80C)	7.6V nominal, XT30	$5.99	HobbyKing (https://hobbyking.com/en_us/turnigy-nano-tech-300mah-2s-35-70c-lipo-pack.html)-class listing, see ../BOM/BOM.xlsx for exact SKU
Payload servo	SG90 9g micro servo	1.8 kg·cm stall torque	~$3.00	AliExpress (bulk-pack pricing, see Payload_Mechanism.md)

Total purchased (Recommended, brushless): see ../BOM/BOM.xlsx, real-priced total is ~ $155 CAD, above my original

$100 CAD target. This is a deliberate, documented trade-off, not a budget overrun I'm glossing over: real brushless FPV micro-motor pricing (~$51 USD for 4 motors, confirmed across both a name-brand retailer and a generic AliExpress listing at nearly identical prices) is simply the real cost floor for hardware that meets the 100-300 g / 20-100 g payload target. I've got room to stretch the budget somewhat, so I'm going with this. The Minimum tier below stays much closer to $100 CAD by accepting reduced payload capacity instead. Swapping the FlySky TX/RX pair for the custom ESP32 link is essentially cost-neutral, actually about $2 CAD cheaper (~$11.67 CAD combined for the RX/TX ESP32s, joysticks, and switches, vs. the ~$13.89 CAD the FlySky receiver alone cost, see ../BOM/BOM.xlsx), it's a scope decision about demonstrating embedded programming, not a budget decision.

1a. Custom RC Link (ESP32, replaces FlySky TX/RX)

I'm replacing the stock FlySky transmitter/receiver pair with a self-written RC link between two ESP32 boards. The reason is deliberate: it turns "buy a receiver" into an actual embedded-programming deliverable (packet protocol, SBUS generation, failsafe logic) that fits the project's stated embedded-systems goal (../Requirements/Requirements.md §1), at close to the same purchased cost as the FlySky pair it replaces. This is a real added engineering/schedule risk, not a drop-in swap — a commercial receiver already has proven failsafe and RF behavior; here I have to build and validate that myself before it's trusted in the air. Flagged honestly, not glossed over.

Link: ESP-NOW (built into the ESP32 WiFi radio, connectionless, no router/pairing needed, low latency, typically ~2-5 ms one-way).

TX firmware (handheld ESP32):

- Reads 4 analog stick axes (throttle, yaw, pitch, roll) + 2 switch inputs (arm, payload release) from GPIO/ADC.
- Packs them into a small struct (channel values + a sequence number + a checksum) and broadcasts it via ESP-NOW at ~50-100 Hz.

RX firmware (onboard ESP32):

- Receives ESP-NOW packets, validates the checksum, updates a local channel-value table.
- Independently, on its own fixed timer (not tied to packet arrival), builds a standard 25-byte SBUS frame from the current channel-value table and writes it out over UART to the FC, at the ~9-14 ms interval Betaflight expects. Decoupling the SBUS timer from RF packet timing keeps the signal to the FC steady even if individual radio packets are late or dropped.
- Failsafe watchdog: if no valid ESP-NOW packet has been received within a timeout (~200-300 ms), the RX firmware sets the SBUS frame's failsafe flag (and/or forces throttle to a safe low value) instead of holding the last good command, the same behavior a commercial receiver provides, but here it has to be explicitly written and bench-tested (Test 5, ../Testing/Test_Plan.xlsx), not assumed.

SBUS signal inversion: standard SBUS is an inverted 100000-baud UART signal. The RX ESP32's UART peripheral can generate this directly (hardware signal-invert option exposed by the Arduino-ESP32 core's HardwareSerial.begin()), so no separate inverter circuit should be needed. Contingency: if the FC doesn't accept the ESP32's inverted signal cleanly, a simple NPN-transistor inverter (~$0.50 CAD in parts, a transistor + two resistors, not separately itemized in the BOM) is a documented fallback.

Known limitation vs. a commercial RC system, stated honestly: ESP-NOW is a WiFi-radio protocol, not a purpose-built RC link. It doesn't have the frequency-hopping/diversity/interference-hardening that AFHDS2A (FlySky) or dedicated systems like ExpressLRS have, and its realistic range with stock antennas is on the order of tens to a couple hundred meters, well short of a commercial system's typical range. That's an acceptable trade for a hobby build flown at close range in a controlled test area, but it's a real limitation, not a hidden one, and range/failsafe behavior must be bench-verified (walk-away failsafe test) before any flight test, not assumed to match FlySky's proven behavior.

2. Minimum-Budget Fallback (Brushed path)

Role	Approach	Why cheaper	Trade-off
FC + motor drivers	A brushed-whoop-class AIO board (built-in motor drivers, no separate ESC)	Comparable price to the brushless FC+ESC combo (~$35-43 USD for real options found), not actually much cheaper, so this isn't the main saving	Similar electronics cost

Role	Approach	Why cheaper	Trade-off
Motors (x4)	Coreless brushed motors (7x16mm or 8.5x20mm class)	~$1-3 each (~$4-12 for 4) vs. ~$51 for brushless, this is the real saving	Much lower thrust; realistic AUW ceiling ~30-50 g, realistic payload capacity only ~5-15 g
Everything else	Same RX/battery/servo approach, scaled down		

The brushed path can realistically land close to $100 CAD (or under, if a receiver/battery is available via university inventory), but it can't meet the 20-100 g payload target. It's included here for completeness and as a fallback, not as the recommendation.

3. Voltage Rails

Rail	Nominal voltage	Source	Loads
Battery / main power	7.6 V (2S HV)	Battery direct	ESC (-> motors), FC power input
5V	5V regulated	FC/ESC board's onboard BEC	RX ESP32 (RC link), payload servo
Signal	3.3V logic (internal to FC)	FC onboard regulator	MCU/gyro internal only

No separate high-current PDB, fuse, or dedicated companion-computer power rail is needed at this scale. The FC/ESC board's onboard BEC handles the onboard RX ESP32 and servo directly (the ESP32-C3 Super Mini's onboard 5V->3.3V regulator accepts the BEC's 5V directly), which is standard practice for a build this small (unlike the archived v1 design's dedicated PDB/UBEC/fuse architecture, which was sized for a much higher-current system). The handheld TX ESP32 is powered separately, off the drone, from a USB power bank.

4. Interfaces

Interface	Between	Protocol
FC <-> ESC	Same board (integrated)	Internal (DShot to onboard MOSFETs)
FC <-> Motors	ESC output -> motor phase wires	3-phase, soldered direct (no bullet connectors needed at this small gauge)
FC <-> RC link	FC <-> onboard RX ESP32	SBUS (single wire + power + ground), generated by custom RX firmware, see §1a
TX ESP32 <-> RX ESP32	Handheld TX <-> onboard RX	ESP-NOW over 2.4GHz WiFi (wireless, no wired connection)
FC <-> Payload servo	FC servo output pad	PWM, mapped to a spare AUX channel in Betaflight
Battery <-> FC/ESC	XT30 connector	Direct, no separate power switch at this current level; unplugging the battery is the standard, sufficient practice for a build this small

5. Wiring Diagram

See Wiring_Diagram.pdf, a simple, clearly labeled diagram (VBAT, 5V, GND, PWM/SBUS). Not a complex multi-page schematic, this is a genuinely simple electrical system.

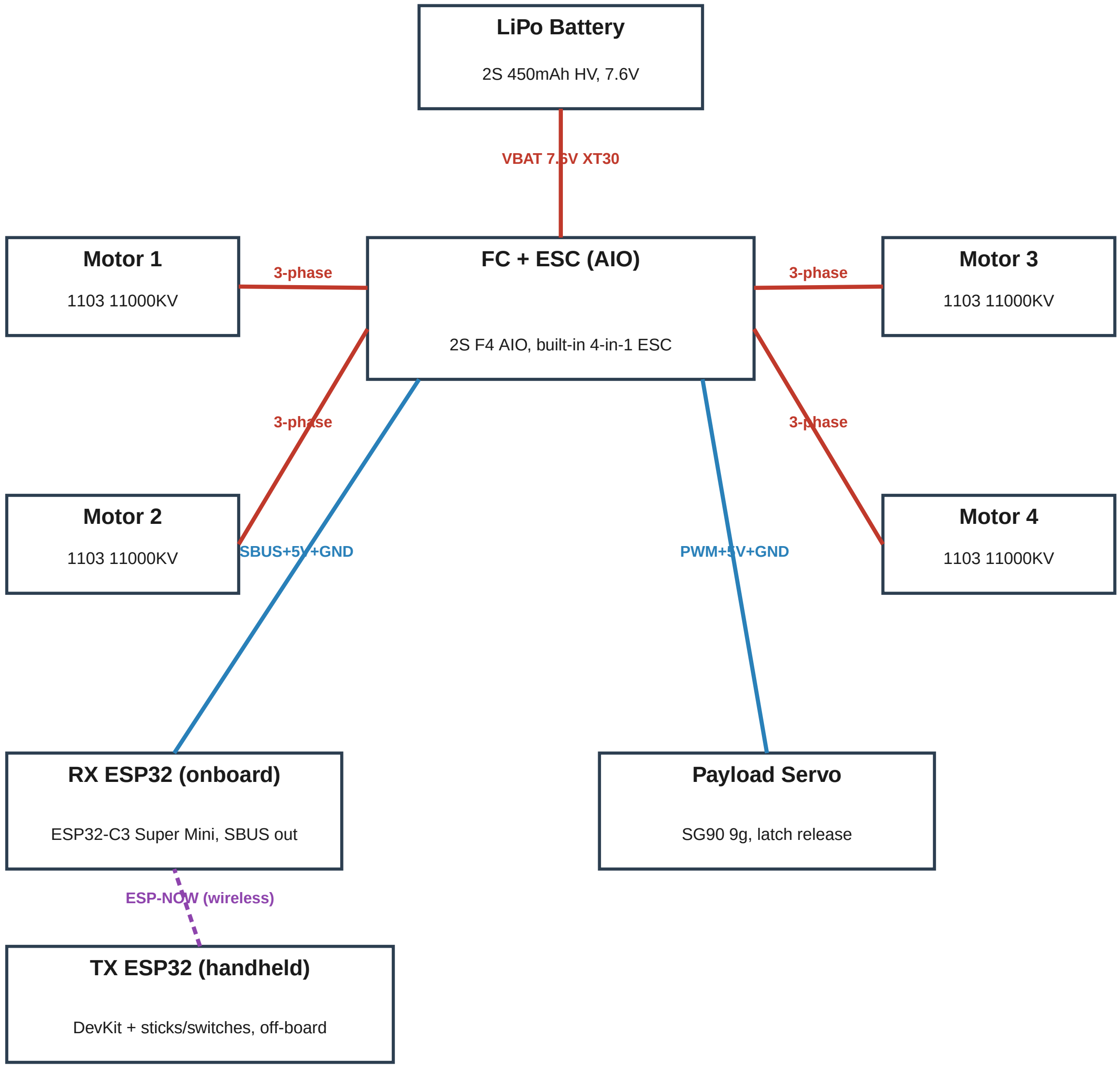
6. Open Items

- Exact battery SKU/mass not independently re-verified this session, confirm at purchase time.
- ESP32 RC-link firmware (TX stick/switch reading, ESP-NOW packet format, RX SBUS encoder, failsafe watchdog) still needs to be written and bench-tested before any motors-on test, see §1a.
- Confirm the ESP32 hardware UART-invert approach actually produces a signal the FC accepts; fall back to the transistor-inverter contingency if not.
- Whether the university's engineering resources include any of the above (battery, ESP32 boards, or even a spare FC/ESC from a robotics club). Every dollar saved here directly closes the gap toward the original $100 CAD target.

SECTION 13

Wiring Diagram

The wiring diagram below shows every component, connector, and signal line in the system, with VBAT/5V/GND/PWM/SBUS clearly labeled per the brief's Section 14. This is a genuinely simple electrical system - one page is enough. The handheld TX ESP32 is off-board and reaches the onboard RX ESP32 wirelessly (ESP-NOW), not by wire, so it is shown separately from the wired onboard system below it.



Legend

- VBAT - battery voltage (7.6V, 2S HV)
- Signal (SBUS, PWM) + regulated 5V from FC/ESC onboard BEC
- ESP-NOW wireless link (2.4GHz WiFi, no physical connection)

GND is common across all wired components (single ground reference). Motor phase wires soldered direct - no bullet connectors needed at this current level. The TX ESP32 is powered separately (USB power bank), not from the drone battery.

SECTION 14

CAD / Manufacturing

Small parts, short print times. This should be a fast, iterative process, not a manufacturing project in its own right.

1. 3D Printing

Parameter	Recommendation	Rationale
Material, frame (arms, hub)	PETG	Better impact resistance than PLA for a part that will occasionally crash; easier than ABS (no enclosure needed); optimizes for strength-to-weight rather than maximum thickness
Material, payload latch/hook	PETG	Repeated-cycle mechanism part; PLA is too brittle for a part that flexes/impacts repeatedly (per the Zbotic reference in Payload_Mechanism.md)
Nozzle	0.4 mm standard	No feature needs finer
Layer height	0.16-0.2 mm	Standard for small structural parts
Wall count	3 perimeters	Enough for a small, lightly-loaded part without adding unnecessary mass. This is a much lighter structural duty cycle than a large UAV frame, so strength-to-weight over thickness applies directly here
Infill	20-25% gyroid	Light but adequate for these small, low-load parts
Print orientation	Arms flat (long axis horizontal), loading direction in-plane with layers, not peeling them apart	Same FDM-anisotropy logic as any printed part
Supports	None needed for the arm/hub geometry as planned; the payload latch mechanism may need minimal supports at the hinge/pivot feature depending on final design	Check once actual CAD geometry exists
Heat-set inserts	M2 brass inserts at the arm-to-hub joint and the servo mount	Required for the modular/repairable design goal (../Research/Concept_Comparison.md)

2. Approximate Print Time & Material Usage (Estimated, no real print exists yet)

Part	Est. print time	Est. material
Hub	~20-30 min	~3-5 g
Arm x4	~10-15 min each (~45-60 min total)	~1-2 g each (~5-8 g total)
Payload mount/latch assembly	~20-30 min	~3-5 g
Handheld TX enclosure (2-piece, holds ESP32 DevKit + sticks + switches)	~30-45 min	~5-10 g
Total	~2-2.75 hours	~15-28 g filament

At typical PETG spool pricing, this is a trivial material cost, well under $5 CAD in filament (confirmed in ../BOM/BOM.xlsx).

3. Machining

None required. This is a deliberate choice (per ../Research/Concept_Comparison.md's selected concept): everything structural is 3D printed, keeping this within the university's basic workshop/3D-printer access without needing CNC/laser-cutting time.

4. Hardware

Item	Spec
Arm-to-hub bolts	M2 socket-head cap screws
Heat-set inserts	M2 brass

Item	Spec
Servo mount bolts	M2
Battery strap	Hook-and-loop (Velcro) strap, or a simple printed friction-fit tray

5. Assembly Sequence

1. Print hub, 4 arms, payload latch assembly, TX enclosure.
2. Install heat-set inserts (soldering iron or dedicated tool, low heat).
3. Bolt arms to hub.
4. Mount motors to arm tips.
5. Mount FC/ESC board to hub, wire to motors (see ../Electrical/Wiring_Diagram.pdf).
6. Mount onboard RX ESP32, battery tray.
7. Mount payload servo + latch assembly.
8. Wire servo to FC/ESC board's servo output.
9. Assemble handheld TX: wire joysticks + switches to the TX ESP32, mount in the printed enclosure.
10. Flash and bench-test the RC-link firmware (TX and RX ESP32) on the bench, off the frame, before wiring the RX into the FC: confirm channel data, AUX switches, and the failsafe watchdog all behave correctly (see ../Electrical/Electrical_Architecture.md §1a).
11. Cable management (zip ties/tape, clear of prop arcs).
12. Continuity check before first power-on (Test 1-3 in ../Testing/Test_Plan.xlsx).

Total build time (printing + mechanical assembly), separate from electronics sourcing/shipping wait time: realistically 2-4 hours of hands-on work. That figure covers fabrication and wiring only; it does not include writing and bench-validating the RC-link firmware (step 10), which is a real software development task with its own separate time cost, not bundled into the "weekend build" estimate. This is still the kind of project that can be finished start-to-finish in a weekend once parts are in hand and the firmware already works, which fits fine within the overall 3-8 week timeline being dominated by parts shipping and iterative testing, not raw build labor.

SECTION 15

Bill of Materials

Full line-item BOM below (also BOM/BOM.xlsx). All prices in CAD, converted from USD at 1 USD = 1.39 CAD (2026-08-30). Items marked CHECK INVENTORY should be excluded from your real out-of-pocket cost if available through university resources.

Category	Part	Qty	Cost (CAD)	Univ.?	Tier
Flight controller	2S F4 AIO FC + Brushless ESC (no RX)	1	$31.96	No	Recommended
RC link	ESP32-C3 Super Mini (onboard RX)	1	$5.55	No	Recommended
RC link	ESP32-WROOM-32 DevKit (handheld TX) + KY-023 dual-axis joystick x2 + momentary/ toggle switch x2	1	$6.12	No	Recommended
Motors	1103 11000KV brushless motor (4-pack)	1	$70.88	No	Recommended
Propellers	Gemfan 65mm 2-blade prop (set of 8, 4 spare)	1	$6.24	No	Recommended
Battery	2S 450mAh HV LiPo, 80C, XT30	1	$8.33	CHECK INVENTORY	Recommended
Servo	SG90 9g micro servo	1	$4.17	No	Recommended
Wiring	Silicone wire, XT30 pigtail, servo extension	1	$6.00	No	Recommended
Connectors	XT30 battery connector (spare/pigtail)	1	$2.00	No	Recommended
Fasteners	M2 socket-head bolts + brass heat-set inserts	1	$10.00	CHECK INVENTORY	Recommended
3D-printing material	PETG filament (frame + payload mechanism + TX enclosure, ~20-30g)	1	$4.00	CHECK INVENTORY	Recommended
Motors (Minimum tier alt.)	Coreless brushed motor 8520 (4pcs + props)	1	$12.51	No	Minimum
Flight controller (Minimum tier alt.)	Brushed whoop AIO FC (built-in motor drivers + RX)	1	$59.76	No	Minimum
Spare/replacement	Spare motor set (4x 1103 11000KV)	1	$70.88	No	Maximum
Spare/replacement	Spare battery (2S 450mAh HV)	1	$8.33	No	Maximum
Tools	LiPo-safe charging bag	1	$12.00	CHECK INVENTORY	Maximum
Tools	2S LiPo balance charger	1	$35.00	CHECK INVENTORY	Maximum

SECTION 16

Cost

Configuration	Total (CAD)	Notes
Minimum	$124.68	Cheapest flying config; brushed motors; cannot meet 20-100g payload target
Recommended	$155.25	Brushless; meets 100-300g mass / 20-100g payload target
Maximum	$281.46	Recommended + spare motor/battery + charger/LiPo bag

Why the Budget Exceeds $100 CAD

Real brushless FPV motor pricing (~$51 USD for a matched set of 4, confirmed at nearly the same price from a name-brand retailer and a generic AliExpress listing) is the real cost floor for hardware meeting the stated mass/payload target - not a shopping oversight. The user has indicated some flexibility to stretch the budget given this. University-inventory availability (battery, hardware, filament) is the main lever to reduce real out-of-pocket cost below the totals above. Swapping the FlySky TX/RX pair for the custom ESP32 RC link (Section 12) is close to cost-neutral, it does not materially change these totals.

SECTION 17

Testing

Full 10-test plan (also Testing/Test_Plan.xlsx). Results are intentionally blank - filled in during actual testing, never pre-filled with assumed outcomes.

Test	Name	Procedure	Success Criteria
Test 1	Frame fit and assembly	Dry-fit all printed parts (hub, arms, payload mount) with hardware before wiring anything	All bolt holes/inserts align; no interference between parts
Test 2	Motor operation	Spin each motor individually via the FC (props OFF), listen/feel for smooth rotation	No grinding, no excess heat, correct direction per Betaflight motor test
Test 3	ESC operation	Verify all 4 ESC channels respond correctly and proportionally to throttle command (props off)	Smooth, proportional response, no channel dropout
Test 4	Gyro/IMU validation	Check Betaflight gyro/accelerometer readings while gently rotating the frame by hand	Readings track rotation correctly, low noise
Test 5	RC-link validation (custom ESP32 link)	Confirm all RC channels respond correctly in Betaflight receiver tab, including the payload AUX channel; then walk the TX out of ESP-NOW range (or power it off) and confirm the RX ESP32's failsafe watchdog triggers correctly within its timeout	All channels move correctly end-to-end, AUX channel triggers servo output, and the FC shows a failsafe condition (not stale/held values) within the RX firmware's timeout when the link is lost
Test 6	Static thrust test	Mount on a simple bench thrust stand (e.g. kitchen scale + fixture), measure thrust (g) vs throttle %	Measured total thrust at a given throttle %; current draw (A)
Test 7	Hover test	Tethered or low-altitude hover in a safe/open area, time the flight to low-battery cutoff	Flight time (min); qualitative stability
Test 8	Payload mechanism test	Bench-test the servo release mechanism repeatedly, without flying, at min/max payload mass	Release success rate over N trials; any binding/ failure
Test 9	Payload flight test	Full mission: take off, fly manually, carry payload, position over target, activate release, land	Payload delivered; release reliability in-flight; landing/frame condition
Test 10	Final demonstration	Complete flight + payload delivery demo, video-recorded for portfolio use	Overall mission success; all measured KPIs compiled

KPI Summary

KPI	Pre-Test Value
Total AUW (bare)	~90 g (Estimated, Frame_Design.md 1)
Total system mass w/ payload	140 g at 50g payload (Estimated)
Required total thrust	~280 gf / 2:1 T/W (Calculated target)
Measured max thrust	NOT YET MEASURED
Current draw (hover)	NOT FOUND / not calculable from available data — measure directly
Flight time	TARGET — no reliable estimate available, see Frame_Design.md 3
Payload capacity (measured)	TARGET — depends on Test 6 thrust result
Payload release reliability	TARGET >= 95% (Test 8)
Frame arm safety factor (simulated)	TARGET >= 3 (FEA_Plan.md)
Print mass (frame + payload mechanism)	~10-18 g (Estimated, Manufacturing_Plan.md 2)
Print time (frame + payload mechanism)	~1.5-2 hours (Estimated)

SECTION 18

Safety

Hazards & Mitigations

- Spinning propellers: always test motors with props OFF first (Tests 2-3); only spin props in a cleared area with no bystanders in the prop plane.
- LiPo battery hazards: charge only in a fireproof LiPo bag, never leave a charging battery unattended, inspect cells for swelling/damage before each use.
- Motor testing: secure the frame before any powered motor test; treat every powered-on vehicle as if props could spin.
- Electrical shorts: continuity-check wiring before first power-on (Test 1); insulate all exposed connections.
- Soldering: standard soldering safety - ventilation, iron rest, eye protection for wire trimming.
- Flight testing: escalate from bench (props off) to tethered/low-altitude to open manual flight; keep an RC kill-switch mapped and tested before every session.
- Payload release: bench-test the release mechanism (Test 8) before any in-flight release test, to confirm it won't release unexpectedly mid-flight.

Emergency Motor Shutdown

The RC transmitter's arm/disarm switch is treated as the emergency stop - tested before every powered session, and the pilot keeps a thumb ready on it during all motor tests and flights. Disarming immediately cuts all motor output regardless of throttle position.

SECTION 19

Development Timeline

Target: 3-8 weeks part-time (NOT 5-8 months, per the corrected project scope).

Week	Focus
Week 1	Research + requirements (this report's Sections 3-6)
Week 2	Component selection + CAD (frame, payload mount)
Week 3	3D printing + electronics sourcing/assembly + RC-link firmware bring-up (ESP-NOW, SBUS encoder, failsafe watchdog)
Week 4	Assembly + bench testing (Tests 1-5, including RC-link range/failsafe validation)
Week 5	Flight testing (Tests 6-7)
Week 6	Payload mechanism + optimization (Tests 8-9)
Week 7-8	Final testing + documentation (Test 10)

Adjust based on parts-shipping time, which will likely dominate the schedule more than hands-on build time (estimated at 2-4 hours for printing/assembly alone, per Mechanical/Manufacturing_Plan.md).

SECTION 20

Resume Value

Resume Project Title

Custom 3D-Printed Micro Quadrotor with Servo-Actuated Payload Release

Resume Bullets

Format: Action + Engineering Method + Technical Detail + Measurable Result. No results are invented, bracketed placeholders mark values that have to come from actual testing (../Testing/Test_Plan.xlsx). Don't fill these in until measured.

1. Designed and fabricated a lightweight, modular, 3D-printed quadrotor frame using CAD and a simple static FEA study, achieving a factor of safety of [X] against a [Y] N motor-thrust load case while keeping bare airframe mass at [Z] g.
2. Selected and integrated a brushless propulsion system (motors, ESC, 2S battery) through first-principles thrust-to-weight calculations, achieving a measured hover flight time of [X] minutes carrying a [Y] g payload.
3. Designed and 3D-printed a custom servo-actuated payload-release mechanism, sized via a torque/moment-arm safety-factor calculation, achieving a [X]% release reliability over [Y] bench trials.
4. Assembled a complete flight-electronics system (flight controller, ESC, custom RC link, battery, payload servo) from individually researched and sourced components, verified via bench continuity/power-on testing with zero electrical faults.
5. Completed a full payload-delivery flight mission (takeoff, manual flight, payload positioning, remote release, landing), demonstrating end-to-end integration of mechanical, electrical, and control-systems engineering on a self-sourced budget of [$X] CAD.
6. Designed and programmed a custom dual-ESP32 RC control link (ESP-NOW wireless protocol, a self-encoded SBUS frame output over UART to the flight controller, and a failsafe watchdog) in C++/Arduino, replacing a stock RC transmitter/receiver pair with self-written embedded firmware, achieving [X] ms link latency and [Y]% failsafe-trigger reliability over [Z] bench trials.

(Six bullets drafted so I can pick the strongest ones once real results exist.)

Why This Project Is Impressive

- It's a complete, small-scale engineering product-development cycle, research, concept comparison, CAD, structural analysis, electronics integration, wiring, manufacturing, and testing, compressed into a project that's actually finishable in weeks, not months. That timeline itself shows good judgment about scope.
- The frame concept (Concept C, modular arm design) was picked over two real alternatives through a documented trade-off analysis (repairability vs. stiffness/mass), not just built as "the obvious quadcopter shape." A genuine, defensible engineering decision and a strong interview story.
- The payload mechanism has a real, calculated safety factor (~9x at worst case, ../Mechanical/Payload_Mechanism.md), showing the mechanical design isn't just "it looks about right."
- The budget research is honest and shows real market awareness: the project openly identifies that real brushless FPV components cost more than a naive $100 target implies, and makes (and documents) a deliberate trade-off instead of hiding the gap.
- The custom ESP32 RC link is real embedded-programming depth, not just mechanical/CAD work: a wireless packet protocol, a from-scratch SBUS encoder, and a failsafe watchdog, deliberately chosen over a cheaper, lower-risk stock receiver specifically to demonstrate that skill (../Electrical/Electrical_Architecture.md §1a).

See Interview_Talking_Points.md for how to defend this project technically.

Interview Talking Points

11 likely technical questions, with guidance on how to answer honestly. Where the answer depends on data not yet measured, don't memorize a fake number, describe the method and cite the real result once testing is complete.

1. "Why this frame design over a standard quadcopter frame?" Walk through the three-concept comparison in ../Research/Concept_Comparison.md. Concept C (modular arm) was chosen specifically for repairability during iterative testing, at a small, deliberate mass/stiffness cost versus a traditional X-frame. Be ready to explain why repairability mattered more than shaving a gram or two for a first build.
1. "How did you size your motors and propellers?" Explain the thrust-to-weight calculation (../Mechanical/Frame_Design.md §2): a bottom-up mass estimate, a 2:1 T/W target, and the resulting required thrust. Be honest that the exact achieved thrust for this specific motor/prop combo wasn't available from a manufacturer source and had to be deferred to bench testing (Test 6). That's a better answer than pretending to have a thrust curve that doesn't exist.
1. "Why did the budget end up above your original $100 CAD target?" This is a genuinely good story, not something to hide: real brushless FPV motor pricing (~$51 USD for a set of 4, confirmed across two independent real listings) is the actual market floor for hardware that meets the payload target, not a shopping mistake. Explain the brushed-motor fallback that was also costed out, and why it wasn't chosen (it can't carry the target payload).
1. "How did you size your payload-release servo?" Walk through the torque/moment-arm calculation in ../Mechanical/Payload_Mechanism.md. The SG90's ~1.8 kg·cm stall torque against a worst-case 100g payload at a 20mm arm gives roughly a 9x safety margin, cross-checked against a real published reference (Zbotic) using the same servo class.
1. "What flight controller firmware are you using, and did you write your own flight-stabilization code?" No, and explain why that's the right call: Betaflight is mature, well-documented, and reimplementing PID stabilization from scratch would be a large time sink for zero real engineering credibility gain on a project this size. The original engineering is in frame design, component integration, wiring, and the payload mechanism.
1. "What was the hardest part of this project?" Should be a real, specific story from the actual build, likely candidates given the design: getting the arm-to-hub joint tolerance right for a snug but removable fit, tuning the payload latch to release reliably without needing excessive servo torque, or a wiring/soldering issue at this small scale. Don't answer this until an actual real problem has been hit and solved.
1. "How did you validate the frame structurally?" Be honest about sequencing: a simple hand-calculated load case first (../Mechanical/Frame_Design.md §4), then one simple static FEA study on the arm (../Mechanical/FEA_Plan.md), deliberately scoped small so FEA doesn't become its own project. If it wasn't run yet before the interview, say so.
1. "What safety precautions did you take?" Propeller/LiPo/electrical hazards, always testing motors with props off first, only spinning props in a cleared area, and a mapped kill-switch/disarm before every flight. Reference ../README.md's safety section.
1. "What would you improve if you built a second version?" Good answers: right-size the motor closer to the exact required thrust once real bench data exists (rather than the estimated target), refine the arm-joint design if bench testing shows it's a stiffness weak point, iterate the payload latch geometry based on release-reliability data from Test 8.
1. "How is this different from just buying a Tiny Whoop and gluing a hook on it?" The frame, motor mounts, and payload mechanism are all original CAD/3D-printed designs with a documented load case and safety-factor analysis, not a repurposed off-the-shelf toy. The component selection (FC/ESC/motor/battery/servo) was individually researched and justified, not copied from one existing kit, and the RC link itself is self-written firmware rather than a bought accessory.
1. "Why build your own RC link instead of just buying a $20 FlySky receiver?" Be honest about the trade: a stock FlySky pair is proven, has built-in failsafe, and is objectively the lower-risk choice for getting a drone in the air. The reason to build a custom ESP32 link anyway is that it turns "buy a part" into an actual programming deliverable, useful for demonstrating embedded/software skills alongside the mechanical work, at close to the same purchased cost (../Design_Journal.md, ../Electrical/Electrical_Architecture.md §1a). Be ready to explain the design: ESP-NOW for the wireless hop, a self-encoded SBUS frame to the FC, and a failsafe watchdog that cuts to a safe state if packets stop arriving, and be honest that its range/interference robustness is weaker than a purpose-built RC system, which is why it's only appropriate for close-range hobby flying, not something safety-critical.

SECTION 21

References

17 unique real sources cited, retrieved via live web search on 2026-08-30.

[1] MakerWorld - https://makerworld.com/en/models/1424214-feather-3-ultra-light-fpv-toothpick-frame
[2] Prusa blog - https://blog.prusa3d.com/how-to-build-a-3d-printed-micro-drone_29310/
[3] RotorBuilds - https://rotorbuilds.com/build/18341
[4] FPV Know-It-All (Joshua Bardwell) - https://www.fpvknowitall.com/fpv-shopping-list-toothpick-fpv-drones-and-parts/
[5] UAVMODEL Insights - https://blog.uavmodel.com/toothpick-and-ultralight-3-inch-fpv-drone-build-guide-2026-parts-list-and-assembly/
[6] FPVCycle - https://fpvcycle.com/products/kabab-toothpick-3-tp3-frame
[7] Printables (Aman_Bhatt) - https://www.printables.com/model/1326088-payload-dropper-mechanism
[8] Hackaday, 2025-01-22 - https://hackaday.com/2025/01/22/diy-drones-deliver-the-goods-with-printed-release/
[9] Zbotic - https://zbotic.in/diy-drone-payload-release-mechanism-step-by-step-build/
[10] BetaFPV - https://betafpv.com/products/1103-brushless-motors
[11] AliExpress - https://www.aliexpress.com/item/32969058968.html
[12] AliExpress - https://www.aliexpress.com/item/1005005319963906.html
[13] AliExpress - https://www.aliexpress.com/item/1005003145871431.html
[14] AliExpress - https://www.aliexpress.com/item/1882818018.html
[15] Pyrodrone - https://pyrodrone.com/products/betafpv-1103-11000kv-2s-brushless-motor-4pcs
[16] GetFPV - https://www.getfpv.com/gemfan-65mm-micro-propellers-1mm-shaft-set-of-8.html
[17] HobbyKing - https://hobbyking.com/en_us/turnigy-nano-tech-300mah-2s-35-70c-lipo-pack.html